

Verification and Validation

Discussion 13 – CSC 4033

Readings

Our reading selections this time are from [Software Engineering: A Modern Approach](#), by Marco Valente, and [Software Engineering at Google](#), by Titus Winters, Tom Manshreck, & Hyrum Wright. Keep in mind while you are reading that some of the learning outcomes may not be directly laid out! Think carefully and build up your answers with the information provided, and as always, feel free to pull in external resources to the discussion.

Required

- 1) [Software Engineering: A Modern Approach, Ch.1.2.4](#) (~1 page)
- 2) [Software Engineering at Google, Ch.11](#) (~22 pages)
- 3) [Software Engineering at Google, Ch.12 – Tests and Code Sharing: DAMP, Not DRY](#) (~8 pages)

Recommended

- The remainder of Ch.11
- Related concepts from other sources we have used already

Learning Outcomes

- Define both verification and validation, explaining how they are complementary but distinct ideas
- Compare the benefits and costs of testing, then discuss whether or not it is worth the time and effort
- Discuss at least five “big picture” project qualities testing seeks to improve (usability, for example)
- List at least three different kinds of tests and discuss when each should be used
- Explain why DRY is often less desirable than DAMP in test code and vice versa in production code